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Calculus - Mistakes

Test 01 (RBR)

Calculus 01:
Functions of a single variable, Limits

The function f : [0, 3] -> [1, 29] defined by f(x) = 2x^3 – 15x^2 + 36x + 1 is

- one-one and onto
- onto but not one-one
- one-one but not onto
- neither one-one nor onto

Full Solution

The function f : [0, 3] -> [1, 29] defined by

$f(x) = 2x^3 - 15x^2 + 36x + 1$

$f'(x) = 6x^2 - 30x + 36$

$= -6(x^2 - 5x + 6)$

$= 6(x - 2)(x - 3)$

f(x) is increasing in [0, 2] and decreasing in [2, 3].

We know, if function is strictly increasing or decreasing in its domain, then it is one-one. But given function is increasing as well as decreasing.

So, f(x) is many one.

f(0) = 1

f(2) = 29 and

f(3) = 28

Range is [1, 29].

The function is onto.

$\lim_{x \to 0}$ (cosec x – cot x)/x is

- 0.5
- 0

$$\lim_{x\to 0} \frac{cosec\ x - cot\ x}{x} = \frac{\frac{1}{sin\ x} - \frac{cos\ x}{sin\ x}}{x}$$

$$= \lim_{x\to 0} \frac{1 - cos\ x}{x\ sin\ x} = \lim_{x\to 0} \frac{1 - 1 + 2\ sin^2(x/2)}{2x\ sin\ (x/2)cos(x/2)}$$

$$= \lim_{x\to 0} \frac{sin\left(\frac{x}{2}\right)}{x\ cos\ \left(\frac{x}{2}\right)} = \lim_{x/2\to 0} \frac{tan\left(\frac{x}{2}\right)}{\frac{x}{2}} \times \frac{1}{2}$$

$$= \frac{1}{2} \times 1 = \frac{1}{2}$$

Time Taken: 193 secs

Let f(x) = $x^{\frac{2}{3}} - 1$ and g(x) = $[(x-1)(x^2+x+1)]^3$

What is fog'(2) equal to ?

- 168
- 308

Full Solution

$g(x) = [(x-1)(x^2+x+1)]^3$

$g(x) = [(x^3-1)]^3$

fog (x) = f(g(x))

$fog\ (x) = ((x^3-1)^3)^{\frac{2}{3}} - 1$

$fog\ (x) = ((x^3-1)^2) - 1$

$fog\ (x) = x^6 + 1 - 2x^3 - 1$

$fog\ (x) = x^6 - 2x^3$

Differentiating the above equation,

$fog'\ (x) = 6x^5 - 6x^2$

At x = 2

$fog'\ (2) = 6(2)^5 - 6(2)^2$

The value of $\lim_{x\to 0} \frac{\tan x - x}{x^2 \tan x}$ is equal to:

O

1/3

1/2

1

$\lim_{x\to 0} \frac{\tan x - x}{x^2 \tan x} = \frac{0}{0}$ is an indeterminate form. Let us simplify and use the L'Hospital's Rule.

$\lim_{x\to 0} \frac{\tan x - x}{x^2 \tan x} = \lim_{x\to 0} \left[\frac{\tan x - x}{x^3} \times \frac{x}{\tan x}\right]$.

We know that $\lim_{x\to 0} \frac{x}{\tan x} = 1$, but $\lim_{x\to 0} \frac{\tan x - x}{x^3}$ is still an indeterminate form, so we use L'Hospital's Rule:

$\lim_{x\to 0} \frac{\tan x - x}{x^3} = \lim_{x\to 0} \frac{\sec^2 x - 1}{3x^2}$, which is still an indeterminate form, so we use L'Hospital's Rule again:

$\lim_{x\to 0} \frac{\sec^2 x - 1}{3x^2} = \lim_{x\to 0} \frac{2\sec x(\sec x \tan x)}{6x} = \lim_{x\to 0} \frac{\sec^2 x \tan x}{3x}$, which is still an indeterminate form, so we use L'Hospital's Rule again:

$\lim_{x\to 0} \frac{\sec^2 x \tan x}{3x} = \lim_{x\to 0} \frac{\sec^2 x \sec^2 x + \tan x[2\sec x(\sec x \tan x)]}{3} = \frac{1}{3}$.

$\therefore \lim_{x\to 0} \frac{\tan x - x}{x^2 \tan x} = 1 \times \frac{1}{3} = \frac{1}{3}$.

Solve: $\lim_{x\to 0}(\sin x)^{\tan x}$

1

Simplify the given limit as follows,

$\lim_{x\to 0}(\sin x)^{\tan x} = k$

Take log both the sides,

$\log [\lim_{x\to 0}(\sin x)^{\tan x}] = \log k$

$[\lim_{x\to 0} \tan x(\log \sin x)] = \log k$

$k = e^{[\lim_{x\to 0} (\tan x)(\log \sin x)]}$

$k = e^{\lim_{x\to 0} \frac{\log \sin x}{\cot x}}$

Since, $\lim_{x\to 0} \frac{log \sin x}{\cot x}$ is of form $\frac{\infty}{\infty}$,

Use the L'hospital rule as follows,

$k = e^{\lim_{x\to 0} \frac{\log \sin x}{\cot x}} = e^{\lim_{x\to 0} \frac{\cot x}{-\text{cosec}^2 x}} = e^0$

$k = e^0 = 1$